

# Marko Cvjetko

PhD Student

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I am a 2nd year PhD student building curiosity-driven AI scientists that autonomously discover and control self-organizing phenomena, combining autotelic RL, vision–language models, and evolutionary search. I am extending this program in two directions: integrating LLMs more deeply as discovery partners in the experimental loop, and testing these methods on more biologically grounded substrates through ongoing collaboration with the [Levin Lab](#) (Tufts). I am a 2026 [DISI](#) fellow and will visit [Clément Hongler's group](#) at EPFL in May 2026.

## PUBLICATIONS

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### Under review

**The Artificial Experimentalist: Discovery and Control of Self-Organizing Phenomena with Autotelic Reinforcement Learning.** M. Cvjetko, B. Hartl, M. Levin, C. Moulin-Frier, P.-Y. Oudeyer. Submitted to *ALIFE 2026*. In collaboration with [Levin Lab, Tufts University](#). [project page]. *Goal-conditioned RL agent that discovers and controls self-organizing patterns in cellular automata via minimal local interventions, with zero-shot generalization to unseen dynamics and action spaces.*

### Journal (invited, in preparation)

Extension of an *ALIFE 2025* paper for the *Artificial Life* journal special issue (~5–8 papers selected from top-ranked submissions and best-paper nominees; 91 accepted at the conference).

### Conference papers

**Exploring Flow-Lenia Universes with a Curiosity-driven AI Scientist: Discovering Diverse Ecosystem Dynamics.** T. Michel, M. Cvjetko (co-first authors), G. Hamon, P.-Y. Oudeyer, C. Moulin-Frier. In *Proc. ALIFE 2025*, pp. 633–643. doi:10.1162/ISAL.a.896. [project page]. *Population-based diversity search (IMGEPS) over system-wide complexity and evolutionary-activity metrics, surfacing ecosystem-level dynamics in Flow-Lenia.*

**Expedition & Expansion: Leveraging Semantic Representations for Goal-Directed Exploration in Continuous Cellular Automata.** S. Khajehabdollahi, G. Hamon, M. Cvjetko, P.-Y. Oudeyer, C. Moulin-Frier, C. Colas. In *Proc. ALIFE 2025*. doi:10.1162/ISAL. [project page]. *Hybrid exploration alternating novelty search with VLM-generated linguistic goals, providing evidence that LLM-generated goals act as stepping stones into unexplored regions of the behavioral space.*

### Workshop paper

**Evolving Symbiosis, from Barricelli's Legacy to Collective Intelligence: a Simulated and Conceptual Approach.** J. Ashford, M. Cvjetko, R. Löffler, B. Sakallıoglu, A. Valerio, M. Tataryn, B. Hartl, L. Pio-Lopez, S. Nichele. Accepted to *GECCO 2026 Workshop Proceedings*. arXiv:2603.08463. Runner-up award at [ALICE 2026 Workshop](#), Copenhagen. *Revisits Barricelli's symbiogenesis simulations as a substrate for studying the emergence of cooperation and collective intelligence.*

### Late-breaking abstract

**Discovering and Controlling Diverse Self-Organised Patterns in Cellular Automata Using Autotelic Reinforcement Learning.** M. Cvjetko, G. Hamon, C. Moulin-Frier, P.-Y. Oudeyer. *ALIFE 2025*, Kyoto, Japan. [HAL]. [project page]

## HONORS, FELLOWSHIPS & RESEARCH VISITS

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### Diverse Intelligences Summer Institute (DISI) Fellow

June 2026

- Selected for a 3-week transdisciplinary residency on the origins, nature, and future of intelligences (cognitive science, AI, biology, philosophy); cohort of ~40 fellows funded by the John Templeton Foundation.

### Funded research visit, group of Clément Hongler (EPFL)

May 2026

- Two-week visit funded by the host group, coinciding with the [Demeco 2026](#) workshop.

### ALICE Workshop Runner-up Award, Copenhagen

Feb 2026

- Awarded for “Evolving Symbiosis, from Barricelli's Legacy to Collective Intelligence”.

## EDUCATION

University of Bordeaux, Inria, FLOWERS team, France

**PhD thesis – Studying the emergence of open-ended evolution in cellular automata using curiosity-driven AI**

Oct 2024 –  
current

- **Supervised by** [Pierre-Yves Oudeyer](#) and [Clément Moulin-Frier](#)
- Applying curiosity-driven AI (evolutionary search, autotelic RL, LLM-driven exploration) to the automated discovery and control of emergent phenomena in Artificial Life systems such as [Lenia](#).
- **Supervision:** co-supervising a Master's intern on the emergence of memory and learning in cellular automata, and two BSc students on a project investigating Turing-completeness in Lenia.
- **Service:** reviewer for *ALIFE 2026*.
- **Collaborations:** ongoing collaboration with Levin Lab (Tufts University) on autotelic agents for the discovery and control of self-organizing patterns; upcoming 2-week funded research visit hosted by [Clément Hongler \(EPFL\)](#), coinciding with the [Demeco 2026](#) workshop (May 2026).

University of Zagreb, Faculty of Electrical Engineering and Computing (FER), Croatia

**MSc, Computer Science**

Sep 2021 – Jul  
2024

- **MSc thesis:** Recognition of the respiratory signal from thermal video.  
**Supervised by** [Prof. Sebastian Haesler](#) and [Prof. Siniša Šegvić](#)  
In collaboration with [Neuro-Electronics Research Flanders](#), project described in [work experience](#) section.
- **Selected courses:** *Machine Learning, Deep Learning, Soft Computing, Parallel Programming*

**Erasmus student exchange at KU Leuven**, Flanders, Belgium

Sep 2022 – Jul  
2023

- Coursework from the [Advanced Master AI](#) programme
- **Selected courses:** *Cognitive Science, Philosophy of Mind, Linguistics and AI Uncertainty in AI, Neural Computing,*

**BSc, Computing**

Sep 2016 – Sep  
2021

- **BSc thesis:** *Music Synthesis using Artificial Intelligence* – LSTM next-note prediction on symbolically represented music. Supervised by [Prof. Domagoj Jakobović](#).

## WORK EXPERIENCE

**Graduate Researcher, [Neuro-Electronics Research Flanders \(imec/KU Leuven/VIB\), Haesler Lab](#)** ([GitHub](#))

Mar 2023 – Jul  
2024

- **Supervised by** [Prof. Sebastian Haesler](#)
- Developed a machine learning framework for extracting and analyzing respiratory responses of head-restrained mice from infrared footage, in collaboration with experimental neuroscientists at Haesler Lab.
- Tackled low-data and domain-shift challenges via unsupervised and self-supervised learning with vision transformers and deep CNNs.
- Configured the data annotation platform, oversaw annotation tasks and trained students on annotation guidelines. Presented in a [poster session](#) at the institute's yearly retreat.

**Machine Learning Engineer Internship, [DoXray](#)**

Jun 2022 – Sep  
2022

- Fine-tuned LayoutXLM — a multilingual multimodal (text + layout + vision) transformer — for named-entity recognition on German invoice documents, benchmarked against a BERT baseline. [Blogpost](#).

## PROJECTS

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Generative models for unsupervised behavioral spaces in open-ended discovery</b>                                                                                                                                                                                                                                                                                                                              | 2026 – current |
| <ul style="list-style-type: none"><li>• Training diffusion models and variational autoencoders to represent complex system behavior and sample their parameters for open-ended diversity search</li><li>• Goal: replace hand-designed semantic descriptors with learned, generative behavior spaces that enable scalable goal-directed search and quality–diversity over continuous cellular automata.</li></ul> |                |
| <b>Music-reactive cellular automata</b>                                                                                                                                                                                                                                                                                                                                                                          | Nov 2024       |
| <ul style="list-style-type: none"><li>• Real-time music-driven Lenia animations: tempo and frequency features drive a continuous cellular automaton (<a href="#">example</a>). Built during the <a href="#">hack1robo</a> hackathon.</li></ul>                                                                                                                                                                   |                |
| <b>Cellular Automata Study in Jax (<a href="#">GitHub</a>)</b>                                                                                                                                                                                                                                                                                                                                                   | Jan – Mar 2024 |
| <ul style="list-style-type: none"><li>• Lenia implementation in Jax; trained a 13-parameter neural network to learn Conway's Game of Life update rule.</li></ul>                                                                                                                                                                                                                                                 |                |
| <b>Automatic Universal Taxonomies for Multi-Domain Semantic Segmentation (<a href="#">GitHub</a>)</b>                                                                                                                                                                                                                                                                                                            | 2023 – 2024    |
| <ul style="list-style-type: none"><li>• Joint label space for semantic segmentation datasets via cross-dataset confusion matrices, enabling to train a single classification head on the union.</li></ul>                                                                                                                                                                                                        |                |
| <b>Can Augmentation Make Better Doctors? Data Augmentation in a Low-Resource Sequence Labeling Task</b>                                                                                                                                                                                                                                                                                                          | 2022           |
| Benchmarked data augmentations for BERT-based sequence labeling on a low-resource medical named entity recognition dataset.                                                                                                                                                                                                                                                                                      |                |

## SKILLS

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|--------------------|------------------------------------------------------------------|
| <b>Programming</b> | Python (Jax, PyTorch, NumPy, Transformers, Optuna, ...), C, Bash |
| <b>Tooling</b>     | Linux, Slurm, Docker, Singularity, Git, LaTeX                    |
| <b>Languages</b>   | Croatian (native), English (fluent), French (intermediate)       |